
Is Perceptual Diminishing Returns Channel-Organized?

A Cone-Alignment Test Against Standard Non-Cone Bases

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Abstract

1 Bujack et al. (2022) showed that perceived color differences violate additivity
2 along the neutral axis—*diminishing returns* (DR)—and concluded that percep-
3 tual color space is not Riemannian. Follow-up work established that DR also
4 occurs on chromatic lines and that its strength varies across color space, but left
5 open whether the nonlinearity is *organized by the cone photoreceptor channels*: a
6 decomposition into per-cone nonlinearities would compactly generate direction-
7 dependent DR everywhere and would localize the compressive mechanism *be-*
8 *fore* the cone signals are combined, whereas alignment with post-receptoral, de-
9 rived quantities would place it *after* aggregation. We test the strongest version
10 of this question supported by public data. Placing four public human dissim-
11 ilarity datasets (the 2022 gray-axis 2AFC triads, the Wiebel–Aguilar–Maertens
12 MLDS lightness data, and the Helm and Ekman hue-circle dissimilarities) into
13 a normalized CIE 2006 cone space, we fit a nested family of city-block dis-
14 tance models—additive (M0), isotropic DR (M1), per-cone DR (M2)—plus a
15 panel of five matched-complexity non-cone controls (M3: opponent, sRGB, CIE
16 RGB, CIELAB, CIE HCL) with per-dataset observation models, and compare
17 them by K -fold cross-validated held-out likelihood with selection-aware boot-
18 strap CIs. The 2022 gray-axis DR finding replicates (held-out $\Delta LL = +574$
19 $[+455, +597]$; subadditivity ratio $S = 0.749 [0.722, 0.793]$ vs. their 0.757). DR
20 is direction-dependent ($M2 - M1 = +220 [+173, +251]$), and a two-channel
21 achromatic/chromatic split already captures most of it ($+174 [+125, +209]$; the
22 residual three-cone gain is $+47 [+36, +56]$, two orders of magnitude smaller than
23 the gaps below). However, **cone alignment is decisively rejected**: per-cone DR
24 loses to the best standard non-cone basis by $-3,840$ held-out log-likelihood (95%
25 CI $[-3,913, -3,809]$; best control: CIE HCL, with CIELAB close), and the three-
26 cone split is not identifiable ($\beta_L - \beta_M$ CI $[-0.34, 0.43]$; the fitted L weight col-
27 lapses to zero). The DR structure aligns with standard *post-receptoral* perceptual
28 axes rather than with cone axes. An explicit hue-locus curvature control con-
29 firms the chromatic DR verdict (DR on arc coordinates: $+191 [+162, +218]$),
30 and near-collinear triad restriction shows the gray-axis DR is robust on the per-
31 ceptual L^* axis ($+454 [+424, +466]$) while the residual cone-axis DR is not
32 ($-17 [-27, +59]$). We conclude that perceptual diminishing returns are channel-
33 organized only at a coarse, post-receptoral granularity and are not specifically
34 cone-aligned: the compressive nonlinearity is organized by derived perceptual
35 quantities, not by the individual cone signals.

1 Introduction

For over a century, perceptual color space has been modeled as a three-dimensional Riemannian manifold whose geodesic lengths equal perceived differences [4, 5, 6]. Bujack, Teti, Miller, Caffrey, and Turton [1] showed that this paradigm fails on its own terms: along the neutral (gray) axis—the one locus on which all classical line elements agree is a geodesic—large differences are perceived as *less* than the sum of small ones. This *diminishing returns* (DR) property, $D(A, B) + D(B, C) > D(A, C)$ for points in order on a geodesic, is incompatible with any Riemannian metric. Their group’s follow-ups extended the finding to two chromatic lines in CIELAB, showing that DR is not constant across color space [2], and formalized a non-Riemannian geometry that accommodates it [3]. The 2025 short paper explicitly poses, but states it cannot test, the natural next hypothesis: that the DR nonlinearity decomposes *per cone channel*.

This paper tests the strongest version of that question the available public data can support. The distinction matters for three reasons.

Compact prediction. If DR decomposes into three per-cone functions f_L, f_M, f_S plus one outer scaling, then a handful of measurements along a few directions generates the diminishing-returns behavior of the *whole* space—a compact generative model—rather than an atlas that must be measured direction by direction.

Mechanism. The alignment localizes where the compression sits in the visual pathway. If DR is aligned with the cones (L, M, S), a compressive nonlinearity is applied to each channel *before* the channels are combined. If it aligns with post-receptoral derived quantities (opponent/luminance channels, or standard perceptual spaces built on them), the compression acts *after* aggregation. This pre- vs. post-combination distinction is a claim about the biology, not a curve-fitting detail.

What a null would mean. If the cone-aligned fit is no better than standard non-cone bases, then DR is not cone-organized, and any compact account must be built on a different axis system—itself an informative outcome. For this reason the cone hypothesis must be tested against a *panel* of popular fixed bases, not one: matching *any* of them falsifies cone-specificity.

Two clarifications keep the claim honest. First, *anisotropy is not the claim*: that DR strength depends on direction is already implied by the non-constancy result of [2], and a direction-dependent fit (M2 below) beating an isotropic one (M1) only restates it. The novel object is *cone-alignment*: whether the per-axis DR structure is specifically tied to the L, M, S cone axes, established by beating the *best* member of a matched panel of fixed non-cone bases on held-out data. Second, the inference that DR contradicts a Riemannian metric requires the stimuli to lie on a geodesic; a curved stimulus locus can mimic subadditivity. We therefore treat the hue-locus curvature confound as logically prior to any chromatic claim and control it explicitly, by data type.

Contributions.

1. **A pre-registered, cross-validated test of cone-alignment** (analysis plan in proposal/analysis_plan.md, written before any fitting). Four public human datasets are placed in a documented, verified CIE 2006 cone space with a stated gauge; a nested model family ($M0$ additive $\subset M1$ isotropic DR $\subset M2$ per-cone DR) is compared against a five-member panel of matched-complexity, fixed non-cone controls ($M3$: opponent, sRGB, CIE RGB, CIELAB, CIE HCL) with per-dataset observation models, identical folds, and selection-aware bootstrap confidence intervals.
2. **An earned null with a positive companion finding.** DR is real (the 2022 gray-axis result replicates in magnitude), direction-dependent, and organized at a coarse-to-intermediate granularity (achromatic vs. chromatic, plus a weak M -vs- S distinction)—but it is *decisively not cone-aligned*: the best standard non-cone bases (CIE HCL, CIELAB) beat the per-cone model by thousands of held-out log-likelihood units, and the finest per-cone (L -vs- M) split is not identifiable in these data.
3. **An explicit identifiability and confound accounting:** collinearity diagnostics and synthetic parameter recovery for the L/M split; a powered cone-vs-nearest-control discrimination check; near-collinear-triad restriction for the gray axis; and an arc-length-vs-chord curvature control for the hue circles, with all verdicts reported before and after control.

2 Methodology

2.1 Cone space, normalization, and gauge

XYZ→LMS. All stimuli are specified by CIE coordinates (see §3). We use the provided CIE 2006 10° cone fundamentals expressed in the CIE 1931 2° XYZ space (`linss2_10e_1.csv`, `cvrl.org`). Because the provided table is per-wavelength while surface/aperture stimuli are given by tristimulus values, we extract the constant 3×3 matrix A by per-cone least squares of the provided fundamentals against the provided 1931 CMFs over each cone’s valid support (390–830 nm for L , M ; 390–615 nm for S):

$$A = \begin{pmatrix} +0.2683 & +0.8466 & -0.0349 \\ -0.3859 & +1.1649 & +0.1029 \\ +0.0267 & -0.0273 & +0.5333 \end{pmatrix}, \quad \text{LMS} = A \text{XYZ}, \quad (1)$$

with relative fit residuals 2.3% (L), 2.3% (M), 13.3% (S). These residuals reflect the known 10°-fundamentals vs. 1931 2°-observer mismatch; they are a documented property of the provided data, not a tuning choice.

Normalization (axis gauge). One global cone frame is used for every dataset and model: $c = (A \text{XYZ}) / (A \text{XYZ}_{D65, Y=100})$ componentwise, i.e., von-Kries axis normalization to the D65 white at $Y=100$, so the D65 white maps to $(1, 1, 1)$ exactly (verified, along with the Illuminant C white, a monochromatic 550 nm reference against the provided fundamental row, and a near-neutral Munsell check; `logs/s1_cone_pipeline.log`). Helm’s Illuminant-C-biased stimuli enter this frame unadapted (a Bradford-adapted variant is a registered sensitivity). An equal-energy normalization is the second registered sensitivity.

Outer/inner gauge. The model family below has gauge freedom between the outer scaling and the per-axis inner functions. We fix it, independently of the data: per-axis reference scales r_k are the RMS of $|\Delta k|$ over all pairs of all datasets (a stimulus-design property), the inner scale is fixed by placing weights on the simplex, and the outer function is confined to the one-parameter concave family $F(y) = \ln(1 + ay)/a$ ($F(0) = 0$, $F'(0) = 1$; $a=0$ gives the identity), with the flexible shape in the per-axis inner functions.

2.2 Model family

Let Δ denote a pair’s coordinate differences in a basis, scaled axis-wise by r_k . The latent distance is

$$D = F\left(\sum_k w_k (|\Delta_k|)^{\beta_k}\right), \quad w_k \geq 0, \sum_k w_k = 1, \beta_k \in [0.2, 1], \quad (2)$$

$F(y) = \ln(1 + ay)/a$, $a \geq 0$. Concavity (DR) corresponds to $\beta_k < 1$ and/or $a > 0$; $F(y) = y$ iff $a = 0$.

- **M0 (additive null):** $\beta_k \equiv 1$, $a = 0$; $D = \sum_k w_k |\Delta_k|$ (2 free weights). Additive, no DR.
- **M1 (isotropic DR):** $\beta_k \equiv 1$, a free (3 free): one direction-independent outer compression of the additive distance—the correction [2] showed is insufficient across two chromatic lines.
- **M2 (per-cone DR):** $\beta_L, \beta_M, \beta_S$ free (6 free). Nests M1 ($\beta_k=1$) and M0 (also $a=0$).
- **M3 (matched non-cone controls, a panel):** the same functional form and 6 free parameters on five fixed, disclosed non-cone bases of the same dimension, each axis RMS-scaled by the same rule so no control carries a conditioning advantage: (i) the post-receptoral *opponent* basis $L-M$, $S-(L+M)$, $L+M$ (a fixed linear map of the normalized cone differences); (ii) *sRGB* ($\text{XYZ} \rightarrow \text{Bradford adaptation to D65} \rightarrow \text{linear sRGB matrix}$); (iii) *CIE RGB* ($\text{XYZ} \rightarrow \text{Bradford adaptation to equal-energy white} \rightarrow \text{CIE 1931 RGB primaries}$); (iv) *CIELAB* (ΔL^* , Δa^* , Δb^* under each dataset’s stated white: D65 for the gray, MLDS, and Ekman data; Illuminant C for Helm); (v) *CIE HCL* ($|\Delta L|$, $|\Delta C|$, wrapped hue difference ΔH in degrees; achromatic stimuli contribute $\Delta C = \Delta H = 0$). No fitted rotations and no data-adaptive linearization: a free rotation would nest M2 and invert the test.

- **Coarse two-channel models** (for the identifiability question): {achromatic $L+M$, chromatic magnitude $\|(L-M, S)\|$ on cones (M2c2-cone; 4 free), and the matched non-cone two-channel control $\{L^*, C^*\}$ from CIELAB (M2c2-lab; 4 free).
- **Curvature baselines** (§4.3): M_{arc} is additive (M0-form) in the shortest hue-arc around the cone-space hue ring plus $|\Delta L^*|$ —an additive metric on a curved locus, i.e., the natural “curvature mimics DR” baseline; M_{arc1} adds the outer DR to the arc coordinates.

2.3 Per-dataset observation models

The four datasets use different response scales and are *not* pooled into one likelihood; they share the latent distance (2) with per-dataset scale/link nuisance parameters:

- **GRAY-2AFC** (Bujack et al. 2022 crowdsourced triads): $P(R=1) = \Phi((D_{\text{ref}, t_2} - D_{\text{ref}, t_1})/\sigma_g)$, aggregated per triad (binomial); nuisance σ_g .
- **WIEBEL-MLDS** (plain viewing condition): the same Thurstonian/Bernoulli form on the two triad intervals, with its own noise σ_w .
- **HELM**: per-subject ratio judgments, $\hat{d}_{s,ij} \sim \mathcal{N}(\alpha_s D_{ij}, \tau_H)$, with per-subject scales α_s (N6a/N6b share one α) and shared τ_H .
- **EKMAN** (aggregate matrix): $\hat{d}_{ij} \sim \mathcal{N}(\alpha_E D_{ij}, \tau_E)$.

Held-out likelihood is the observation-model log-likelihood of held-out rows; the combined metric is the sum over datasets (identical held-out rows for every model).

2.4 Fitting, cross-validation, bootstrap

Fits are multi-start L-BFGS-B (structured start + random restarts, seeds logged; convergence checked by agreement across starts) with nuisance parameters profiled in closed form (ratio datasets) or by a fast 1-D search (2AFC σ). Cross-validation uses fixed, seed-documented, basis-independent folds, identical for every model: GRAY-2AFC 10 triad-disjoint folds; WIEBEL-MLDS leave-one-observer-out (10); HELM 9 pair-disjoint folds (all subjects in every fold, so per-subject scales are always estimable from training pairs); EKMAN 10 pair-disjoint folds. For each fold, the model is trained on everything except that fold and evaluated on the fold, so each held-out observation contributes exactly once to every model’s total.

Bootstrap ($B=200$ per fold): training data are resampled by the appropriate units (triads for GRAY, observers for WIEBEL/HELM, pairs for EKMAN); models are refit; and the nuisance vector is re-profiled on the full training set at the refit parameters before held-out evaluation (without this, observation groups absent from a resample would receive degenerate nuisance values). The cone-alignment headline is *selection-aware*: the M2–M3 gap is recomputed as the paired minimum over the panel within every bootstrap draw, so the best control is re-selected inside each resample (no winner’s curse).

2.5 Identifiability and confound-control methods

Identifiability. (i) Collinearity diagnostics of the cone-difference design (correlations, condition number, VIFs, effective rank; pooled and chromatic-only). (ii) Synthetic parameter recovery: responses simulated at the exact real stimulus design and trial counts from the fitted M2 and from the fitted M3-opponent (the nearest control, whose rotation from the cones lies largely in the poorly sampled $L \approx M$ subspace), 100 seeds each, clearly labeled as synthetic. (iii) Cone-vs-control discrimination power: held-out model selection on the synthetic replicates. (iv) The powered channel-level comparison the data support: the coarse two-channel cone model vs. M1 and vs. the matched two-channel non-cone control.

Confound control. Triad data: the neutral axis is a straight line in every linear frame, so hue-locus curvature is structurally absent; the control is the *near-collinear-triple restriction* to the 290/320 gray triads whose reference lies strictly between the two tests (MLDS triads are ordered by construction). Hue-circle data: the *arc-length-vs-chord* control—an additive metric on the hue arc (M_{arc}) vs. chord-based DR, with DR-on-arc (M_{arc1}) testing whether DR survives when the geometry is allowed to be the curvature.

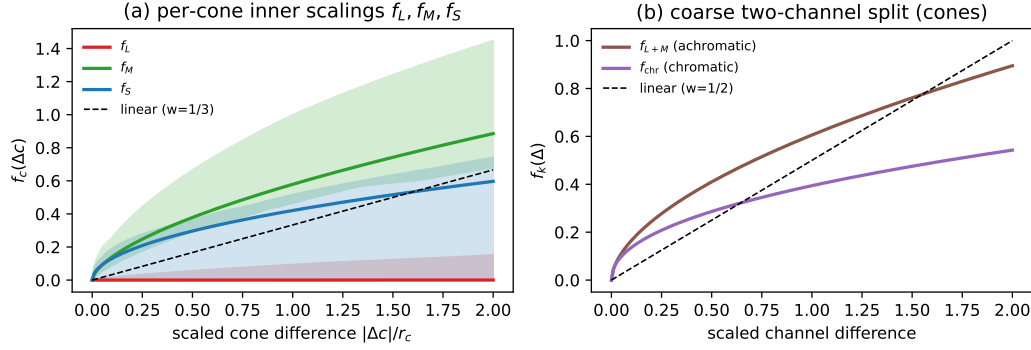


Figure 1: Fitted inner scalings with full-data bootstrap bands ($B=200$; $\log s/s3$). (a) Per-cone f_L, f_M, f_S in the normalized cone frame: the fitted L -cone weight collapses to zero ($w_L = 0.017$, CI $[0.000, 0.094]$) and β_L is unidentifiable (CI $[0.21, 0.99]$); β_M and β_S are mildly distinct ($\beta_M - \beta_S = +0.11$, CI $[+0.02, +0.34]$). (b) The coarse two-channel split (achromatic $L+M$ vs. chromatic magnitude): both channels are strongly compressive.

3 Experimental Setup

Datasets and licenses. Only the provided public data are used. (1) **GRAY-2AFC**: 91,896 responses over 320 unique triads on the neutral L^* axis ($L_s \in \{30, \dots, 70\}$, tests 0–100), from the `lanl/color` repository (BSD-3). (2) **WIEBEL-MLDS**: the MLDS experiment of Wiebel, Aguilar & Maertens [11] (OSF, CC BY 4.0), plain viewing condition (12,000 trials, 10 observers over 10 reflectance levels); stimulus luminances (cd/m^2 , deterministic) are read from the matching files of the same release, and the achromatic stimuli are placed at D65 chromaticity. (3) **HELM**: 10 Munsell Value-6 tiles under Illuminant C, pairwise ratio-scaled dissimilarities for the 11 color-normal subject rows of Helm [9] (via `smacof : : helm`, GPL-3; colorimetry from Helm’s Table I; color-deficient rows excluded from the primary analysis). (4) **EKMAN**: 14 spectral colors (434–674 nm) at equal brightness, aggregate 14×14 dissimilarities [10] (via `smacof : : ekman`, GPL-3; spectral-locus chromaticities from the provided CIE 1931 CMFs). Stimulus placement: $L^* \rightarrow \text{XYZ}$ (D65) for the gray data; measured luminances (D65 chromaticity) for WIEBEL; $x, y, Y \rightarrow \text{XYZ}$ (Illuminant C) for HELM; $x, y, Y \rightarrow \text{XYZ}$ (D65 convention) for EKMAN. All converted cone coordinates are logged.

Preprocessing. 2AFC responses are aggregated to per-triad counts (identical likelihood). HELM subject N6’s two sessions share one scale parameter. Reference scales r_k are fixed from the pooled stimulus design (§2.1).

Gray-axis replication summary. The 2022 finding is reproduced three ways (`logs/s2_gray_replication.log`): the consensus regression (their Eq. 10) gives $\beta_{\bar{d}} = -0.0061$, $t(28) = -5.73$, $p < 10^{-5}$ (2022: -0.005 , $t(60) = -5.57$; our grouping yields 32 ($\bar{d}, \Delta d$) cells vs. their 64, a grouping difference only); M1 beats M0 on held-out triad-disjoint folds by $+573.6$ LL (95% CI $[+454.7, +596.9]$); and the subadditivity ratio on the L^* axis, $S = D(30\text{-step})/2D(15\text{-step})$ at mid-gray, is 0.749 (95% CI $[0.722, 0.793]$), matching the 2022 report ($2.06/2.72 = 0.757$). The independent WIEBEL-MLDS suprathreshold check gives $+210.1$ ($[+173.6, +229.1]$).

4 Experimental Results

4.1 Model comparison

Table 1 and Figures 1–2 give the primary comparison. Three regimes are visible. (i) *DR exists and replicates*: M1 beats M0 on the gray data by $+573.6$ $[+454.7, +596.9]$, with matching magnitude ($S_{L^*} = 0.749$ vs. 2022’s 0.757), and the WIEBEL-MLDS data independently confirm ($+210.1$). (ii) *DR is direction-dependent*: M2 beats M1 by $+220.4$ $[+172.9, +251.3]$, and the coarse two-channel cone model already captures most of this ($+173.5$ $[+125.1, +208.5]$ over M1); the full

Table 1: Primary results. Held-out log-likelihood (LL) totals over all four datasets with identical folds; Δ LL rows are differences with selection-aware bootstrap 95% CIs ($B=200$ per fold). Every value traces to proposal/logs/ (s2, s3, s4, s5; consolidated in s7).

Quantity	Estimate	95% CI	Comparison	Notes
<i>Gray-axis DR replication (anchor)</i>				
Gray M1–M0 held-out	+573.6	[+454.7, +596.9]	(2022: concave f)	$P(\Delta > 0) = 1.00$
magnitude S (L^* axis)	0.749	[0.722, 0.793]	2022: 0.757	cone: 0.57 [0.56, 1.00]
WIEBEL M1–M0 held-out	+210.1	[+173.6, +229.1]	independent	suprathreshold
<i>Held-out fit (combined, all datasets)</i>				
M0 (additive)	−70,737.8			2 free params
M1 (isotropic DR)	−70,013.7			3 free params
M2 (per-cone DR)	−69,793.3			6 free params
M3 opponent	−69,775.7			matched complexity
M3 sRGB	−69,815.9			matched complexity
M3 CIE RGB	−69,785.9			matched complexity
M3 CIELAB	−66,076.0			matched complexity
M3 CIE HCL	−65,953.7			matched complexity
M2c2 (cone 2-ch.)	−69,840.2			4 free params
M2c2 (L^* , C^* 2-ch.)	−66,320.2			4 free params
M2 – M1 (anisotropy)	+220.4	[+172.9, +251.3]		$P(\Delta \leq 0) < 10^{-3}$
M2 – best M3 (cone-alignment)	−3839.6	[−3913, −3809]	best: CIE HCL	signif. negative
<i>Identifiability</i>				
Granularity: $\beta_L - \beta_M$	+0.01	[−0.34, +0.43]		L/M not separable
M2c2-cone – M1 (channel org.)	+173.5	[+125.1, +208.5]		coarse split wins
M2 – M2c2-cone	+46.9	[+35.5, +56.4]		residual M -vs- S gain
M2c2-cone – M2c2-lab	−3520.0	[−3590, −3496]		non-cone 2-ch. wins
Cone-vs-opponent discrimination	§4.2		SYNTHETIC	powered
<i>Hue-curvature confound (before/after)</i>				
Chromatic M1 – M0 (chords)	+155.0	[+136.9, +173.7]	before	DR present
M_arc1 – M_arc (arcs)	+191.4	[+162.3, +217.9]	after	survives on arcs
M_arc – M1	−229.9	[−244.8, −206.1]	after	arc baseline loses
M2 – best ctrl. (chromatic, arc incl.)	−125.5	[−135.2, −108.8]	after	best: CIE HCL
Gray restricted (cone basis)	−17.2	[−27.4, +59.5]	after	not robust
Gray restricted (L^* axis)	+454.3	[+424.3, +465.7]	after	DR robust

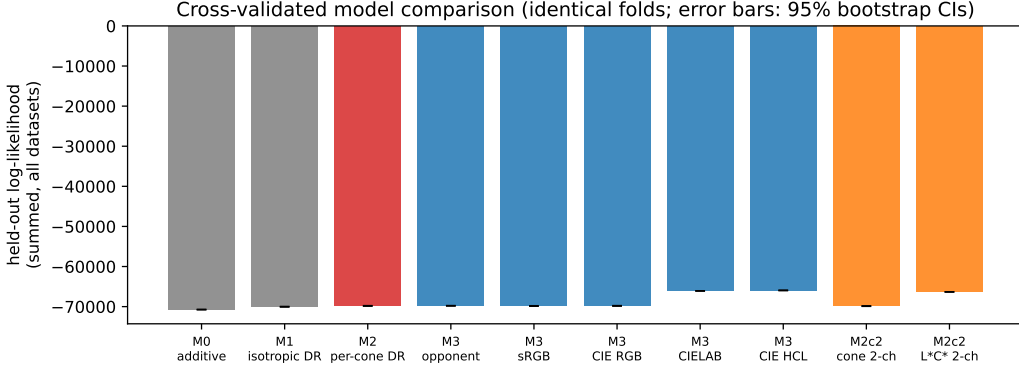


Figure 2: Cross-validated model comparison (identical folds; error bars: 95% bootstrap CIs). The per-cone model M2 is indistinguishable from the physical non-cone bases (opponent, sRGB, CIE RGB) but is beaten by the perceptually-defined bases CIELAB and CIE HCL by $\sim 3,800$ held-out LL.

three-cone form adds a small residual gain (+46.9 [+35.5, +56.4] over the two-channel model), attributable to the M -vs- S distinction. The fitted per-cone exponents are strongly compressive ($\beta_M = 0.60$ [0.40, 0.78], $\beta_S = 0.49$ [0.31, 0.62]) with a strong outer compression ($a = 36.2$ [23.7, 52.9]). (iii) *Cone-alignment is rejected*: M2 shows no meaningful separation from the physical non-cone bases—the gaps are small with mixed signs (opponent -17.6 [−71.8, +12.5]; sRGB +22.6 [+15.3, +29.0]; CIE RGB -7.4 [−12.4, −1.2]), and two of three point estimates favor the control—while the perceptual bases, CIELAB and CIE HCL, win by 3,700–3,840 held-out LL, with the selection-aware M2–best-M3 interval [−3,913, −3,809] excluding zero by thousands of units. The two-channel non-cone control (L^* , C^*) shows the same pattern (−3,520 [−3,590, −3,496] vs. its cone counterpart).

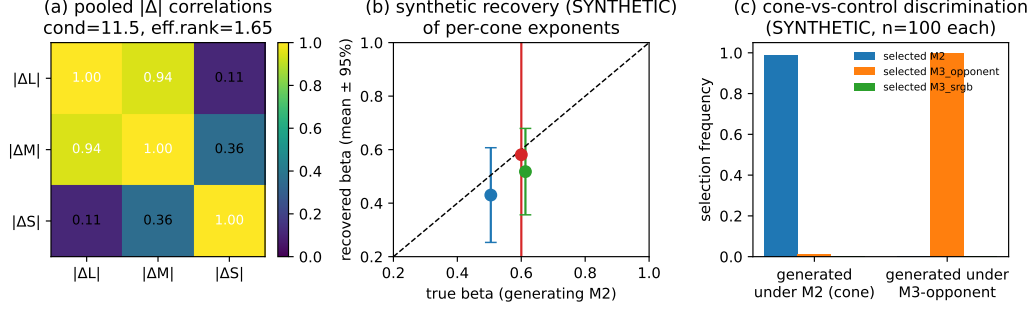


Figure 3: Identifiability. (a) Correlations of the pooled $|\Delta L|$, $|\Delta M|$, $|\Delta S|$ design with condition number and effective rank. (b) Synthetic parameter recovery (SYNTHETIC data, simulated at the real design from the fitted M2): per-cone exponents; the L -cone exponent is not recoverable. (c) Cone-vs-control discrimination power on synthetic replicates generated under M2 and under M3-opponent.

223 Per-dataset held-out totals (logs/s3) locate the CIELAB/HCL advantage: it is large on both gray-
 224 axis datasets (e.g., GRAY: M3_hcl $-59,881$ vs. M2 $-62,783$; WIEBEL: $-5,020$ vs. $-5,848$), where
 225 the engineered L^* axis matches the lightness data almost perfectly, and persists on both hue circles
 226 (HELM: $-1,112$ vs. $-1,173$; EKMAN: $+59.7$ vs. $+10.5$).

227 4.2 Identifiability

228 **Collinearity.** On the gray and MLDS datasets the cone differences lie on a single ray (the achro-
 229 matic direction), so those datasets are perfectly collinear by construction—they anchor one combi-
 230 nation of the cone axes and nothing more. On the chromatic data, $\text{corr}(|\Delta L|, |\Delta M|) = 0.804$ and
 231 $\text{corr}(|\Delta M|, |\Delta S|) = 0.885$, the signed chromatic difference correlation is $\text{corr}(\Delta L, \Delta M) = 0.527$
 232 (pooled, including the achromatic datasets: 0.935), the chromatic-only design has effective rank
 233 1.07 of 3, and the second-moment matrix of the signed chromatic differences has eigenvalues
 234 $(103.3, 0.045, 3 \times 10^{-5})$: the sampled difference directions are dominated by a single composite
 235 direction (within 2.3° of a cone axis), with the remaining two directions barely sampled (logs/s4).
 236 The hue circles therefore nearly confound the per-axis structure in exactly the way anticipated.

237 **Synthetic recovery (SYNTHETIC).** Simulating from the fitted M2 at the exact real design and
 238 noise levels (100 replicates), the per-cone exponents recover as β_M (mean 0.52, sd 0.08) and β_S
 239 (mean 0.43, sd 0.09), but β_L is not recoverable (mean 0.58, sd 0.25), the fitted L weight collapses
 240 toward zero (mean 0.023), and the outer compression a has a shallow profile (mean 18.2 vs. true
 241 36.3, sd 9.9)—the finest per-cone split is not identifiable at these directions, and the outer/inner
 242 curvature trade is only weakly resolved.

243 **Discrimination power (SYNTHETIC).** With data generated under M2, the held-out protocol se-
 244 lects M2 over M3-opponent and M3-sRGB in 99/100 replicates; with data generated under M3-
 245 opponent it selects M3-opponent in 100/100 (per-replicate held-out deltas $+13.3$ [$+2.1, +26.3$] and
 246 -21.3 [$-33.3, -10.8$] respectively). The design *does* have power to discriminate cones from the
 247 nearest control at the sampled directions, so the near-tie between M2 and the opponent/sRGB/CIE-
 248 RGB controls on real data is an earned indistinguishability, not a power failure—and the real-data
 249 CIELAB/HCL gaps ($\sim 3,800$) are two orders of magnitude beyond this resolution scale.

250 **Resolvable granularity.** Combining the evidence: the resolvable structure sits *between* the coarse
 251 two-channel split and the full three-cone split. The two-channel achromatic/chromatic model cap-
 252 tures most of the anisotropy ($+173.5$ over M1), and the three-cone form adds a small, signifi-
 253 cant residual ($+46.9$ [$+35.5, +56.4$]), consistent with the one per-cone distinction the bootstrap
 254 resolves ($\beta_M - \beta_S = +0.11$ [$+0.02, +0.34$]). But the L -cone channel is not separable ($\beta_L - \beta_M$
 255 CI $[-0.34, +0.43]$; $w_L \rightarrow 0$; synthetic β_L unrecoverable), and the two-channel non-cone control
 256 (L^*, C^*) beats its cone counterpart by 3,520. The channel organization that survives is therefore
 257 coarse-to-intermediate and post-receptoral, not cone-specific.

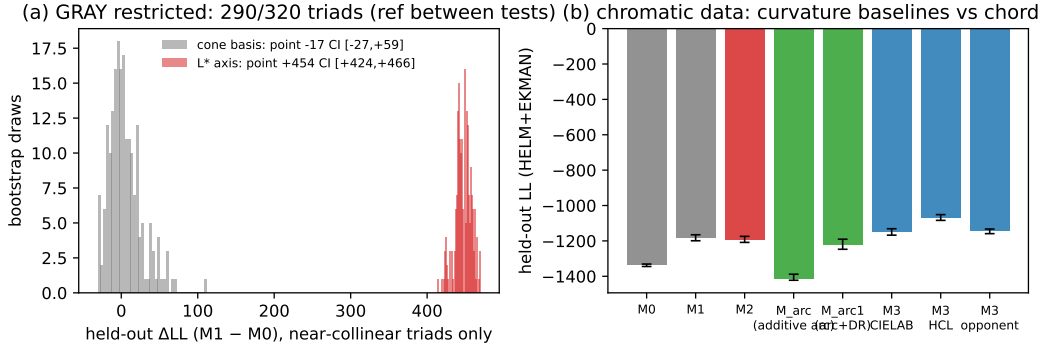


Figure 4: Confound controls. (a) Gray-axis near-collinear-triple restriction: on the 290 triads whose reference lies between the tests, the L^* -axis DR verdict is robust (+454 [+424, +466]) while the residual cone-axis DR is not (−17 [−27, +59]). (b) Chromatic arc-vs-chord: DR survives on arc coordinates (+191 [+162, +218]) and the additive arc baseline is *worse* than chord DR (−230 [−245, −206]), so hue-locus curvature does not explain the chromatic DR; the cone-alignment rejection persists with arc baselines included.

4.3 Confound control (before/after)

Gray triads. All gray stimuli lie on a straight line in every linear frame, so curvature is structurally absent; the near-collinear-triple restriction tests robustness of the DR verdict to keeping only the 290/320 triads in which the reference lies strictly between the tests. The L^* -axis DR (the 2022-style analysis) is robust (+454.3 [+424.3, +465.7]); the residual DR on the cone achromatic axis is not (−17.2 [−27.4, +59.5]). The reason is instructive: the concave $L^* \rightarrow Y$ (cone) transform already absorbs the DR-shaped compression on between-triads—on that subset the additive cone model and the cone DR model fit identically (the fitted $a \rightarrow 0$)—so the cone-axis DR signal is carried by the 30 non-between triads. This is a substantive limitation for reading the gray-axis evidence in cone coordinates and is disclosed as such.

Hue circles. Before control, DR on the chromatic data is +155.0 [+136.9, +173.7] (M1 vs. M0 on cone chords). After control: DR on arc coordinates is +191.4 [+162.3, +217.9] (M_arc1 vs. M_arc), and the additive arc baseline is *worse* than chord DR by 229.9 [206.1, 244.8]—curvature alone cannot explain the data. The cone-alignment verdict is unchanged with the arc baselines added to the control set (M2 − best control = −125.5 [−135.2, −108.8], best: CIE HCL).

4.4 Sensitivities and orientation supplement

All registered sensitivities preserve every verdict (logs/s6): equal-energy cone normalization (M2−M1 +220.6; M2−opponent −20.5); HELM Bradford-adapted to D65 (+227.1; −23.3); β range widened to [0.2, 2] (M2’s held-out fit is unchanged at −69,793, the optimum being interior; CIELAB and HCL improve by ≤ 30 and remain $\sim 3,700$ ahead). The free-orientation supplement (a fitted 3-angle Givens rotation of the cone frame, in which M2 is nested; not part of the M2-vs-M3 verdict) fits angles (−0.013, +0.030, −0.236) rad with all three bootstrap CIs including zero—the best aggregation axes are not reliably rotated away from the cone axes; the failure of cone-alignment is a failure of the *axis metric and its per-axis nonlinearities*, not of a rotation the cone basis happened to miss.

5 Analysis

Verdict (a): is DR direction-dependent? Yes, but modestly and mostly at coarse granularity. The per-cone model beats isotropic DR by +220 held-out LL; the two-channel achromatic/chromatic split captures most of it (+174 over M1), and the residual three-cone gain (+47 [+36, +56]) is two orders of magnitude smaller than the non-cone gaps below.

Verdict (b): is DR specifically cone-aligned? No—an earned and decisive null. The per-cone model does not beat the *best* standard non-cone basis; it loses to CIE HCL and CIELAB by $\sim 3,800$ held-out LL (CI excluding zero by three thousand units), and it does not even separate reliably from the opponent, sRGB, or CIE RGB bases. Because the synthetic discrimination check shows the design *can* separate cone-generated from opponent-generated data at these directions, the tie among physical bases and the loss to perceptual bases are findings about the data, not about power.

Verdict (c): at what granularity is the structure identifiable? Not the full three-cone split. The L -cone weight collapses to zero, β_L is unrecoverable in synthetic experiments, and the $\beta_L - \beta_M$ interval covers zero; the one per-cone distinction the data resolve is β_M vs. β_S (and the matching small $+47$ held-out gain of M2 over the two-channel cone model). The resolvable granularity is therefore an achromatic-vs-chromatic split plus an intermediate M -vs- S distinction—an empirically post-receptoral decomposition, not a per-receptor one.

Mechanism reading (calibrated). The pre-combination mechanism—a compressive nonlinearity applied to each cone signal before the channels are combined—is rejected by these data. What the data support is compression organized on standard post-receptoral perceptual axes: CIELAB’s (L^* , a^* , b^*) and its cylindrical form. We emphasize the limit of this reading: CIELAB axes are themselves nonlinear (cube-root) functions of cone excitations, so “per-axis DR on CIELAB” is a statement about the effective axis system of the compression, not a pinned physiological site; in particular the CIELAB advantage partly reflects that its L^* axis already implements much of the lightness compression that the cone-achromatic model must otherwise learn (see the gray-axis restriction analysis, §4.3). What can be said sharply is the negative: the individual cone channels do not carry separable DR nonlinearities that generate the whole space compactly.

Failure analysis and limitations. (i) Point held-out totals are best-of-multi-start and can be mildly optimistic per model; the small physical-basis gaps ($|\Delta| \leq 72$ LL) should not be over-read, and all verdicts rest on the bootstrap CIs. (ii) The residual cone-axis gray DR is not robust to the near-collinear restriction; the CIELAB advantage on the gray/WIEBEL datasets is dominated by the engineered L^* axis. (iii) The S-cone row of the XYZ-space fundamentals fits a constant matrix worst (13% residual), a property of the provided observer data that propagates to all cone-axis results; the non-cone bases avoid it. (iv) M0/M1/M2 use an L^1 (city-block) aggregation; the object of study is the outer/inner nonlinearity within an additive-aggregation family, and the L1-vs-L2 aggregation choice is not tested (a limitation inherited from the registered design). (v) The datasets barely overlap in cone-space direction (the gray data anchor the achromatic axis, the hue circles the chromatic plane); cross-dataset identification of the shared latent form is correspondingly weak, which is part of the identifiability story rather than a failure. (vi) The Ekman aggregate matrix is a single averaged response surface; its Gaussian observation model is an approximation. (vii) β_{C^*} of M2c2-lab sits at the lower bound (0.2) in some fits—a boundary solution we report as is.

6 Related Work

Bujack et al. (2022) [1] established DR on the neutral axis with a Thurstonian 2AFC analysis and concluded perceptual color space is not Riemannian. We reproduce their finding in magnitude, extend the axis system to a normalized CIE 2006 cone frame and to three further datasets, and—going beyond replication—use the cone frame to ask the organizational question their analysis could not.

Stark, Turton & Bujack (EuroVis 2025) [2] extended DR to two chromatic lines and showed its strength is not constant across color space, posing the per-cone-decomposition hypothesis and stating it cannot be tested on two geodesics. Our contribution begins exactly there: with four datasets spanning the achromatic axis and two saturation levels of the hue circle, we test channel organization against a panel of matched non-cone controls. Confirming that DR is direction-dependent (our M2/M2c2 vs. M1) restates their non-constancy in our setting; the new results are the decisive rejection of cone-specificity and the identification of the resolvable two-channel granularity.

Bujack et al. (CGF 2025) [3] formalize a non-Riemannian geometry that can accommodate DR. Our results inform what such a geometry’s coordinates should be: the compressive structure is organized on post-receptoral perceptual axes, not on cone axes.

Classical and modern antecedents. Judd’s “principle of diminishing returns” [7], MacAdam’s $p < 1$ exponents [8], Helm’s logarithmic transform [9], and Izmailov & Sokolov’s spherical embedding [12] all anticipated nonadditivity; our gray-axis magnitude ($S \approx 0.75$ on L^*) sits squarely in that tradition. The observation that CIELAB-like axes out-perform physical trichromatic axes for suprathreshold scaling echoes why CIELAB was standardized [13]—but the failure of the *per-cone* decomposition, and the two-channel granularity result, are new.

7 Conclusion

We asked whether the diminishing-returns nonlinearity of perceptual color difference is organized by the cone photoreceptor channels. On four public human datasets placed in a verified, normalized CIE 2006 cone space, with a pre-registered nested model family, a matched-complexity panel of five fixed non-cone controls, per-dataset observation models, selection-aware bootstrap CIs, an explicit identifiability analysis, and curvature controls appropriate to each data type, the answer is a well-powered **no**: DR is real (the 2022 gray-axis finding replicates in magnitude), it is direction-dependent, and it is organized at a coarse-to-intermediate achromatic/chromatic (+ weak M -vs- S) granularity—but it is not specifically cone-aligned, losing to standard post-receptoral perceptual bases (CIE HCL, CIELAB) by thousands of held-out log-likelihood units, with the finest per-cone split unidentifiable in these data. Compact models of diminishing returns should therefore be built on derived perceptual axes rather than on per-cone nonlinearities, and mechanistic accounts should look past the photoreceptor outputs for the site of the compression. The main limitations—sparse direction sampling, a single averaged response surface for one dataset, and the L1 aggregation family—map directly onto the data collection that would tighten the granularity claim: many directions, per-trial responses, sampled densely in the chromatic plane.

Reproducibility statement

All analysis code (proposal/code/), logs and machine-readable results (proposal/logs/), figures (proposal/figures/), and the pre-registered analysis plan (proposal/analysis_plan.md, time-stamped before the fitting code was written) are included in the submission. Every quantitative claim in this paper traces to a log entry; seeds for all stochastic operations are fixed and logged; all synthetic data are labeled as such and were used only for parameter-recovery and power checks.

Data provenance and licensing

GRAY-2AFC: lanl/color repository, BSD-3. WIEBEL-MLDS: Wiebel, Aguilar & Maertens (2017), OSF, CC BY 4.0. HELM and EKMAN dissimilarity matrices: redistributed via the smacof R package (GPL-3); stimulus colorimetry from Helm (1964), Table I, and the CIE 1931 spectral locus (physical facts). CIE 2006 fundamentals and CIE 1931 CMFs: cvrl.org under its citation terms. No new human data were collected.

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